

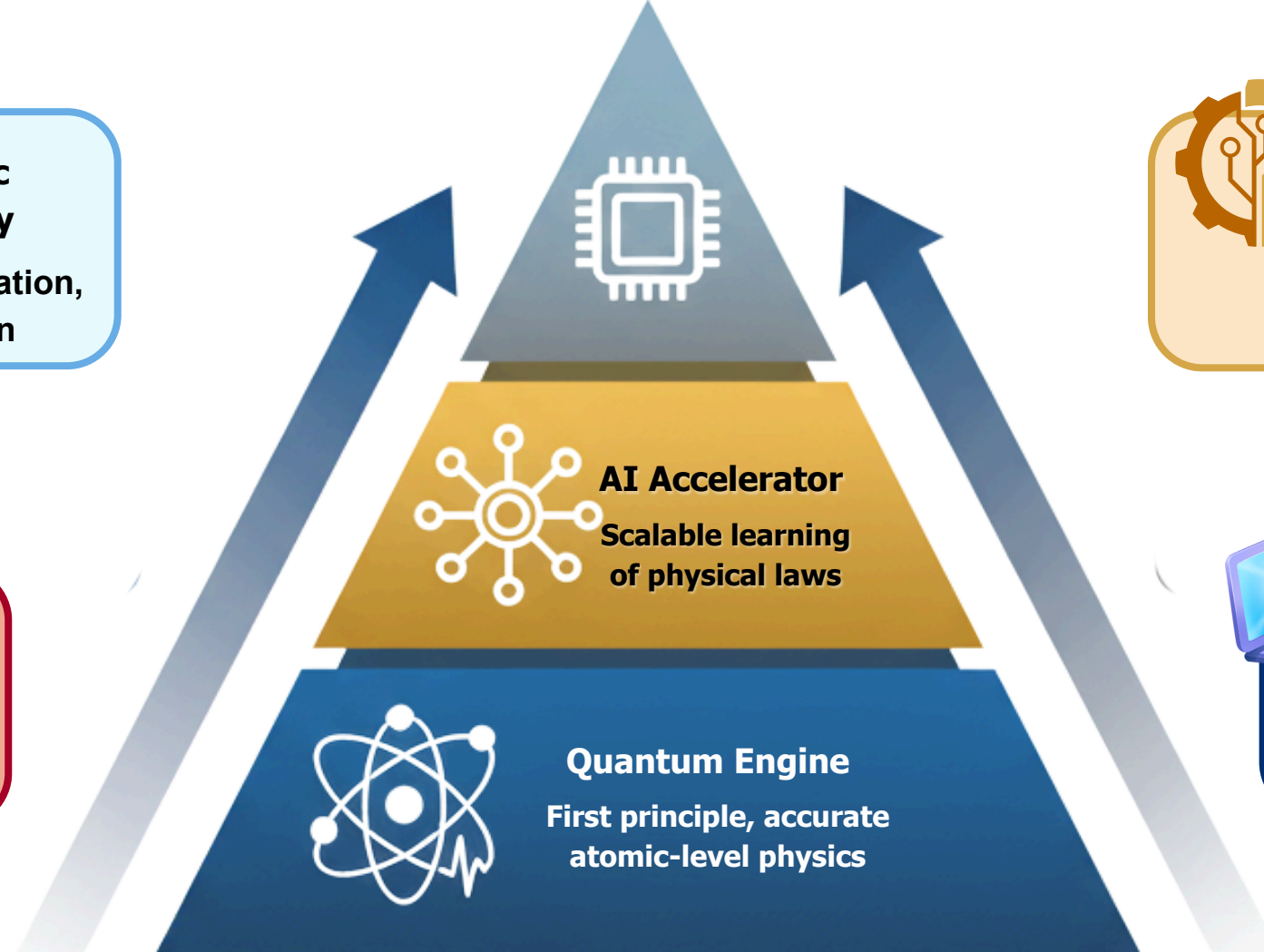
Digital Twins
Predictive reliability
modeling



**Scientific
Discovery**
Programmable degradation,
atomic stabilization



**PHM
Innovations**
Real-time twins,
PoF modeling



**Technological
Impact**
Reliable 2D devices,
faster discovery



Methological
Physics–AI integration,
multiscale bridging